

A MULTIPROCESS STATE SPACE MODEL FOR MODELLING THE RATE OF RETURN ON ORDINARY STOCK

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Key words: Kalman filter; market model; multiprocess state space model.

Summary: A multiprocess state space model that models the rate of return on stocks is developed. As such, the return generating process is assumed to be in one of four possible regimes, with a corresponding regime indicator probability value being updated recursively, as new data becomes available.

1. INTRODUCTION

In mean variance portfolio theory, a single factor model of the form

$$(1.1) \quad r_t = \alpha + \beta m_t + e_t$$

where r_t denotes the rate of return on a stock in period t , m_t , the rate of return on a market index in period t , e_t , a random error term that is assumed to be uncorrelated with the market index in period t and α and β , parameters that are unique to the security, is often used to describe the correlation structure that exists between a security's return and that of a suitably chosen market index. Based on the observed notion that when the return on a market index begins to increase in value, the return on certain stocks will also tend to increase in value, the above equation simply separates the return on a stock into an expected component

$$\mathcal{E}(r_t) = \alpha + \beta \mathcal{E}(m_t)$$

and an associated risk component that is measured by

$$\text{var}(r_t) = \beta^2 \text{var}(m_t) + \text{var}(e_t).$$

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Known as the beta coefficient for the above stock, initial attempts at estimating β have generally attempted to use ordinary least squares techniques. The implementation of such a procedure can, however, only be justified if the beta coefficient does not change over time, and if the returns, r_t and m_t , come from a bivariate normal distribution. Past studies by Mandelbrot (1963), Fielitz and Smith (1972), Chang and Weiss (1991) and Blume (1971), have tended to question both these assumptions, and as a result these authors have generally attempted to incorporate some form of time varying parameter estimation technique into their model structure or to assume that the rate of return on the stock has a stable distribution. Our approach will be to develop a multiprocess state space model for predicting stock market returns. By specifying a mixture of normal distributions to cope with the non-normality that has been observed in the past stock market returns, and by using the Kalman filtering equations to update one's parameter estimates as new data become available, it is hoped that a more realistic and flexible pricing model for stock returns will result.

2. THE STATE SPACE AND MULTIPROCESS STATE SPACE MODEL REPRESENTATIONS

A state space model representation is defined by the following system of equations, viz.

$$\begin{aligned} \mathbf{y}_t &= \mathbf{F}_t \mathbf{x}_t + \mathbf{v}_t \\ \mathbf{x}_t &= \mathbf{G}_t \mathbf{x}_{t-1} + \mathbf{w}_t \\ \mathbf{x}_0 &\sim N(\hat{\mathbf{x}}_0|_0, \Sigma_0|_0) \end{aligned}$$

where

$\mathbf{y}_t = n \times 1$ vector of process observations that have been collected at time point t

$\mathbf{x}_t = p \times 1$ random vector of unobservable process parameters (state variables)

$F_t = n_t \times p$ measurement equation matrix assumed to be known at the time point t

$G_t = p \times p$ system equation matrix assumed to be known at the time point t

$\{v_t\}$ = a sequence of uncorrelated normal random vectors with zero mean and variance-covariance matrix $E\{v_t v_t'\} = R_t$

$\{w_t\}$ = a sequence of uncorrelated normal random vectors with zero mean and variance-covariance matrix $E\{w_t w_t'\} = Q_t$

The model description is then completed by assuming that v_t, w_t are independently distributed of x_{t-1} , given the past observations $\{y_1, \dots, y_{t-1}\}$.

Given that the process parameter values $\{Q_t, R_t, F_t, G_t, \Sigma_0|_0, \hat{x}_0|_0\}$ are known, recursive equations for estimating x_t from a set of observations $\{y_1, \dots, y_t\}$ can be given by

$$\hat{x}_t|t = \hat{x}_t|t-1 + \Sigma_t|t-1 F_t' (F_t \Sigma_t|t-1 F_t' + R_t)^{-1} e_t$$

$$\Sigma_t|t = \Sigma_t|t-1 - \Sigma_t|t-1 F_t' (F_t \Sigma_t|t-1 F_t' + R_t)^{-1} F_t \Sigma_t|t-1$$

$$\hat{x}_t|t-1 = G_t \hat{x}_{t-1}|t-1$$

$$\Sigma_t|t-1 = G_t \Sigma_{t-1}|t-1 G_t' + Q_t$$

$$e_t = y_t - F_t \hat{x}_t|t-1$$

where

$$\hat{x}_t|t \equiv E\{x_t | y_1, \dots, y_t\}$$

$$\Sigma_t|t \equiv E\{(x_t - \hat{x}_t|t)(x_t - \hat{x}_t|t)' | y_1, \dots, y_t\}$$

It is, however, possible that the global behaviour of a time series process may not be adequately described by a *single* state space model structure of the form that is given above. Rather, by varying the choice of parameter specifications for the quadruple $\{F_i, G_i, R_i, Q_i\}$, the resulting model structures (or a suitably chosen linear combination of them) may at times provide a better representation of the evolutionary pattern of the time series process. Known as a multiprocess state space model representation, Harrison and Stevens (1976) have developed a class of model structures where the uncertainty regarding the evolutionary change in the process is characterized, for each time period point, by a set of state space models of the form

$$\begin{aligned} y_t &= F_t x_t + v_t^{(i)} & v_t^{(i)} &\sim N(0, R_t^{(i)}) \\ x_t &= G_t x_{t-1} + w_t^{(i)} & w_t^{(i)} &\sim N(0, Q_t^{(i)}) \\ i &= 1 \dots M \end{aligned}$$

where each of the M different model structures is assumed to model a different type of change that can occur in the process. Letting $\{S_t = i\}$ denote a model indicator variable for the i -th model structure, and

$$\begin{aligned} \hat{x}_t^{(i)} &= E\{x_t | y_1 \dots y_t, S_t = i\} \\ \Sigma_t^{(i)} &= \text{var}\{x_t | y_1 \dots y_t, D_t = i\} \end{aligned}$$

denote a conditional mean vector and variance covariance matrix for the i -th model structure, a composite estimator of the form

$$E\{x_t | y_1 \dots y_t\} = \sum_{i=1}^m p_t^{(i)} \hat{x}_t^{(i)}$$

is developed for x_t where

$$p_t^{(i)} = P(S_t = i | y_1 \dots y_t)$$

denotes the conditional probability that model i is operating at the time period point t .

The formulation and derivation of the multiprocess dynamic linear model is well known (Harrison and Stevens (1976)). Briefly, by letting

$$p_{t-1}^{(i)} \equiv P(S_{t-1} = i | \mathbf{y}_1 \dots \mathbf{y}_t)$$

denote the conditional probability that model i is in operation during the time period $t-1$, a joint probability-density function of the form

$$\mathbf{x}_{t-1} | \mathbf{y}_1 \dots \mathbf{y}_t \sim \sum_{i=1}^M p_{t-1}^{(i)} N(\hat{\mathbf{x}}_{t-1}^{(i)} | \Sigma_{t-1}^{(i)})$$

is updated to form a new joint probability-density function of the form

$$\mathbf{x}_t | \mathbf{y}_1 \dots \mathbf{y}_t \sim \sum_{i=1}^M \sum_{j=1}^M p_t^{(i,j)} N(\hat{\mathbf{x}}_t^{(i,j)} | \Sigma_t^{(i,j)})$$

as a new observation $\mathbf{y}_t$ becomes available. The probabilities

$$p_t^{(i,j)} \equiv P(S_{t-1} = i, S_t = j | \mathbf{y}_1 \dots \mathbf{y}_t)$$

are then, assuming $\{S_t = i\}$ and $\{S_u = j\}$ are independent for $t \neq u$, calculated via a direct application of Bayes's theorem, yielding

$$p_t^{(i,j)} \propto L(\mathbf{y}_t | S_{t-1} = i, S_t = j, \mathbf{y}_1 \dots \mathbf{y}_{t-1}) p_{t-1}^{(i)} \pi_t^{(j)}$$

where

$$L(\mathbf{y}_t | S_{t-1} = i, S_t = j, \mathbf{y}_1 \dots \mathbf{y}_{t-1}) \sim N(\mathbf{F}_t' \mathbf{G} \hat{\mathbf{x}}_{t-1}^{(i)} | \mathbf{F}_t' (\mathbf{G} \Sigma_{t-1}^{(i)} | \mathbf{G}' + \mathbf{Q}_t^{(j)}) \mathbf{F}_t' + \mathbf{R}_t^{(j)})$$

and

$$\pi_t^{(j)} \equiv P(S_t = j)$$

denotes the unconditional probability that model i is in operation during the time period point t . The process parameters, $\hat{\mathbf{x}}_t^{(i,j)}$ and $\Sigma_t^{(i,j)}$, are then recursively updated using the following Kalman filtering equations

$$\begin{aligned}
\hat{\mathbf{x}}_t^{(i,j)} &= \mathbf{G}\hat{\mathbf{x}}_{t-1}^{(i)} + \mathbf{K}_t^{(i,j)} \mathbf{e}_t^{(i,j)} \\
\Sigma_t^{(i,j)} &= \Sigma_t^{(i,j)}|_{t-1} - \mathbf{K}_t^{(i,j)} \mathbf{F}_t' \Sigma_t^{(i,j)}|_{t-1} \\
\Sigma_t^{(i,j)}|_{t-1} &= \mathbf{G}\Sigma_{t-1}^{(i)} \mathbf{G}' + \mathbf{Q}_t^{(j)} \\
\mathbf{K}_t^{(i,j)} &= \Sigma_t^{(i,j)}|_{t-1} \mathbf{F}_t' (\mathbf{F}_t \Sigma_t^{(i,j)}|_{t-1} \mathbf{F}_t' + \mathbf{R}_t^{(j)})^{-1} \\
\mathbf{e}_t^{(i,j)} &= \mathbf{y}_t - \mathbf{F}_t \mathbf{G}\hat{\mathbf{x}}_{t-1}^{(i)}|_{t-1}
\end{aligned}$$

The following collapsing procedure which minimizes the Kulback-Leibler distance measure for the above normal mixture (Pena and Guttman (1989)) is then employed to reduce the model complexity of the multiprocess dynamic linear model to an order M , viz.

$$\begin{aligned}
p_t^{(j)} &= \sum_{i=1}^M p_t^{(i,j)} \\
\hat{\mathbf{x}}_t^{(i)}|_t &= \left[\sum_{i=1}^M p_t^{(i,j)} \hat{\mathbf{x}}_t^{(i,j)} \right] / p_t^{(j)} \\
\Sigma_t^{(j)}|_t &= \sum_{i=1}^M p_t^{(i,j)} \left[\Sigma_t^{(i,j)} + (\hat{\mathbf{x}}_t^{(i,j)} - \hat{\mathbf{x}}_t^{(j)}|_t)(\hat{\mathbf{x}}_t^{(i,j)} - \hat{\mathbf{x}}_t^{(j)}|_t)' \right] / p_t^{(j)}
\end{aligned}$$

with

$$\mathbf{x}_t | \mathbf{y}_1 \dots \mathbf{y}_t \sim \sum_{j=1}^M p_t^{(j)} \mathcal{N}(\hat{\mathbf{x}}_t^{(j)}|_t, \Sigma_t^{(j)}|_t).$$

Application of the above results, however, require that one has knowledge of the observation and system equation error variance-covariance matrix parameters. When they are not known, one can employ an approach that was developed by Smith and West (1983), where, on letting $\lambda^{-1} = \sigma^2$,

they assumed that

$$\text{var}(\mathbf{v}_t^{(i)}) = \lambda^{-1} \mathbf{R}^{(i)}$$

$$\text{var}(\mathbf{w}_t^{(i)}) = \lambda^{-1} \mathbf{Q}^{(i)}$$

where $\mathbf{R}^{(i)}$ and $\mathbf{Q}^{(i)}$ are known, and then made the additional assumptions that

$$\mathbf{x}_0 \sim N(\hat{\mathbf{x}}_0|_0, \lambda^{-1} \mathbf{C}_0)$$

and

$$\lambda | \mathbf{y}_1 \dots \mathbf{y}_{t-1}, S_{t-1} = i \sim G(\tfrac{1}{2} \delta_{t-1}, \tfrac{1}{2} \rho_{t-1}^{(i)})$$

where $G(\delta, \rho)$ denotes a gamma distribution, with parameters δ and ρ respectively. Letting

$$\lambda \sim G(\tfrac{1}{2} \delta_0, \tfrac{1}{2} \rho_0)$$

denote an initial distribution for λ , then, on defining,

$$\mathbf{x}_{t-1} | \mathbf{y}_1 \dots \mathbf{y}_{t-1}, S_{t-1} = i \sim N(\hat{\mathbf{x}}_{t-1}^{(i)} |_{t-1}, \lambda^{-1} \mathbf{C}_{t-1}^{(i)} |_{t-1})$$

one can obtain the results that

$$\mathbf{x}_t | \mathbf{y}_1 \dots \mathbf{y}_t, \lambda, S_{t-1} = i, S_t = j \sim N(\hat{\mathbf{x}}_t^{(i,j)} |_{t-1}, \lambda^{-1} \mathbf{C}_t^{(i,j)} |_{t-1})$$

$$\lambda | \mathbf{y}_1 \dots \mathbf{y}_t, S_{t-1} = i, S_t = j \sim G(\tfrac{1}{2} \delta_t, \tfrac{1}{2} \rho_t^{(i,j)})$$

where

$$\hat{\mathbf{x}}_t^{(i,j)} |_{t-1} = \mathbf{G} \hat{\mathbf{x}}_{t-1}^{(i)} |_{t-1} + \mathbf{K}_t^{(i,j)} \mathbf{e}_t^{(i,j)}$$

$$\mathbf{C}_t^{(i,j)} |_{t-1} = \mathbf{C}_{t-1}^{(i,j)} - \mathbf{K}_t^{(i,j)} \mathbf{F}_t' \mathbf{C}_{t-1}^{(i,j)}$$

$$\mathbf{P}_t^{(i,j)} = \mathbf{F}_t \mathbf{C}_{t-1}^{(i,j)} \mathbf{F}_t' + \mathbf{R}^{(j)}$$

$$\delta_t = \delta_{t-1} + n$$

$$\rho_t^{(i,j)} = \rho_{t-1}^{(i)} + \mathbf{e}_t^{(i,j)'} (\mathbf{P}_t^{(i,j)})^{-1} \mathbf{e}_t^{(i,j)}$$

Furthermore, on noting that

$$\mathbf{y}_t | \mathbf{y}_1 \dots \mathbf{y}_{t-1}, \lambda, S_{t-1} = i, S_t = j \sim N(\mathbf{F}_t \mathbf{G} \hat{\mathbf{x}}_{t-1}^{(i)} |_{t-1}, \lambda^{-1} \mathbf{P}_t^{(i,j)})$$

and

$$\lambda | \mathbf{y}_1 \dots \mathbf{y}_{t-1}, S_{t-1} = i, S_t = j \sim G(\frac{1}{2} \delta_{t-1}, \frac{1}{2} \rho_{t-1}^{(i)})$$

the result can be obtained that, unconditional on λ ,

$$\mathbf{y}_t | \mathbf{y}_1 \dots \mathbf{y}_{t-1}, S_{t-1} = i, S_t = j \sim T(\mathbf{F}_t \mathbf{G} \hat{\mathbf{x}}_{t-1}^{(i)} |_{t-1}, \frac{\rho_{t-1}^{(i)}}{\delta_{t-1}} \mathbf{P}_t^{(i,j)})$$

a multivariate T distribution with degrees of freedom, δ_{t-1} , mode,

$\mathbf{F}_t \mathbf{G} \hat{\mathbf{x}}_{t-1}^{(i)} |_{t-1}$, and positive definite scale matrix, $\frac{\rho_{t-1}^{(i)}}{\delta_{t-1}} \mathbf{P}_t^{(i,j)}$, respectively.

Also

$$p_t^{(i,j)} \propto \left[\delta_{t-1} + \frac{\delta_{t-1}}{\rho_{t-1}^{(i)}} \mathbf{e}_t^{(i,j)'} (\mathbf{P}_t^{(i,j)})^{-1} \mathbf{e}_t^{(i,j)} \right]^{-\frac{\delta_t}{2}} p_{t-1}^{(i)} \pi_t^{(j)}$$

and a conditional density for $\mathbf{y}_t$ can be given by

$$f(\mathbf{y}_t | \mathbf{y}_{t-1}, \dots, \mathbf{y}_1) = \sum_{i=1}^M \sum_{j=1}^M \pi_t^{(j)} p_{t-1}^{(i)} T\left[\mathbf{F}_t \mathbf{G} \hat{\mathbf{x}}_{t-1}^{(i)} |_{t-1}, \frac{\rho_{t-1}^{(i)}}{\delta_{t-1}} \mathbf{P}_t^{(i,j)}\right].$$

The following collapsing procedure (West, Harrison and Migon (1985)) can then be employed to yield

$$\mathbf{x}_t | \mathbf{y}_1 \dots \mathbf{y}_n, S_t = j \sim N(\hat{\mathbf{x}}_t^{(i)} |_t, \lambda^{-1} \mathbf{C}_t^{(j)})$$

$$\lambda | \mathbf{y}_1 \dots \mathbf{y}_n, S_t = j \sim G(\frac{1}{2} \alpha_t, \frac{1}{2} r_t^{(i)})$$

where

$$\begin{aligned}\hat{\mathbf{x}}_t^{(j)} &= \left[\sum_{i=1}^m p_t^{(i,j)} \mathbf{x}_t^{(i,j)} \right] / p_t^{(j)} \\ \mathbf{C}_t^{(j)} &= \sum_{i=1}^m p_t^{(i,j)} \left[\mathbf{C}_t^{(i,j)} + (\hat{\mathbf{x}}_t^{(i,j)} - \hat{\mathbf{x}}_t^{(j)})(\hat{\mathbf{x}}_t^{(i,j)} - \hat{\mathbf{x}}_t^{(j)})' \right] / p_t^{(j)} \\ (r_t^{(j)})^{-1} &= \left[\sum_{i=1}^m p_t^{(i,j)} / r_t^{(i,j)} \right] / p_t^{(j)}\end{aligned}$$

3. A MULTIPROCESS MODEL FOR PRICING STOCKS

Turning our attention to the development of a suitable multiprocess model for the return on a stock, let us assume that the rate of return on a stock can be adequately represented by a model of the form

$$\begin{aligned}r_t &= \alpha_t + \beta_t m_t + \delta r_t^{(j)} \\ \alpha_t &= \alpha_{t-1} + \delta \alpha_t^{(i)} \\ \beta_t &= \beta_{t-1} + \delta \beta_t^{(i)}\end{aligned}$$

where $\delta r_t^{(i)}$, $\delta \alpha_t^{(i)}$ and $\delta \beta_t^{(i)}$ denote a set of suitably chosen random error terms with

$$\begin{aligned}\text{var}(\delta r_t^{(i)}) &= \lambda^{-1} \sigma_r^{2(i)} \\ \text{var}(\delta \alpha_t^{(i)}) &= \lambda^{-1} \sigma_\alpha^{2(i)} \\ \text{var}(\delta \beta_t^{(i)}) &= \lambda^{-1} \sigma_\beta^{2(i)} \\ \text{cov}(\alpha_t^{(i)}, \beta_t^{(i)}) &= 0\end{aligned}$$

Then, by specifying, a priori, the following set of values for the above variance-covariance parameters, viz.

	$\sigma_r^{2(i)}$	$\sigma_\alpha^{2(i)}$	$\sigma_\beta^{2(i)}$
$j = 1$	≈ 1	≈ 0	≈ 0
$j = 2$	≈ 1	large	≈ 0
$j = 3$	≈ 1	≈ 0	large
$j = 4$	large	≈ 0	≈ 0

the following localized patterns of change in the evolution of the process can be modelled, viz.

- (i) a steady state (no change in the evolution of r_t)
- (ii) a change in level in the evolution of the parameters α_t
- (iii) a change in level in the evolution of the parameters β_t
- (iv) a transient (temporary) change in the evolution of r_t

For the purposes of making a comparison, we will also consider the development of a localized linear growth model for the rate of return on a stock where it is assumed that

$$\begin{aligned}
 r_t &= \mu_t + \delta r_t^{(i)} & \delta r_t^{(i)} &\sim N(0, \lambda^{-1} \sigma_r^{2(i)}) \\
 \mu_t &= \mu_{t-1} + \gamma_t + \delta \mu_t^{(i)} & \delta \mu_t^{(i)} &\sim N(0, \lambda^{-1} \sigma_\mu^{2(i)}) \\
 \gamma_t &= \gamma_{t-1} + \delta \gamma_t^{(i)} & \delta \gamma_t^{(i)} &\sim N(0, \lambda^{-1} \sigma_\gamma^{2(i)})
 \end{aligned}$$

The parameters, μ_t and γ_t , can be viewed as modelling the actual level of the rate of return on the stock and the incremental growth in this level at time t , respectively. By specifying the following set of parameter values for the variance-covariance terms

	$\sigma_r^{2(i)}$	$\sigma_\mu^{2(i)}$	$\sigma_\gamma^{2(i)}$
$j = 1$	≈ 1	≈ 0	≈ 0
$j = 2$	≈ 1	large	≈ 0
$j = 3$	≈ 1	≈ 0	large
$j = 4$	large	≈ 0	≈ 0

the following types of changes can be modelled, namely

- (i) a steady state (no change in the evolution of r_t)
- (ii) a change in the level of the rate of return on the stock
- (iii) a change in the slope of the level of the rate of return on the stock
- (iv) a temporary change in the rate of return on the stock.

Clearly one could also consider combinations of these above four states. For example, letting $\sigma_r^2 = 1$, and setting large values for σ_μ^2 and $\sigma_{\gamma'}^2$, a simultaneous change in the level and slope can be modelled. We will not, however, consider the effect of such a state in this study.

4. RESULTS

Monthly stock price data from the internationally traded diamond company De Beers was chosen for our study. The Johannesburg Stock Exchange All Actuaries Index was used as a proxy for the market index and the period of study was taken from January 1980 to June 1991. In order to consider both the estimation and forecasting performance of our two state space models, estimates for the unknown model parameters, $\{\alpha_p, \beta_p, p_t\}$ and $\{\mu_p, \gamma_p, p_t\}$, were computed (see figures 1 and 2), and then used to compute the following one-step-ahead forecasts for y_p , namely

$$\hat{y}_t|_{t-1} \equiv \mathcal{E}\{y_t|y_{t-1}, \dots, y_1\} = \sum_{i=1}^M \sum_{j=1}^M \pi_t^{(j)} p_{t-1}^{(i)} \mathbf{F}_t \mathbf{G} \hat{\mathbf{x}}_{t-1}^{(i)}|_{t-1}.$$

The performance of these one-step-ahead forecasts are given in figure 3, where, for both state space model structures, the one-step-ahead prediction errors, defined by

$$\hat{e}_t|_{t-1} = \mathbf{y}_t - \hat{\mathbf{y}}_t|_{t-1}$$

are plotted.

Noting that

$$\hat{\lambda}_T = \mathcal{E}\{\lambda | \mathbf{y}_T, \dots, \mathbf{y}_1\} = \frac{\delta_T}{\rho_T}$$

defines a time varying estimate for the inverse of the variance on the rate of return on the stock, a plot of the precision parameter estimates for both state space model structures is also given in figures 1 and 2. This in itself is a useful result in that shifts in the variance of stock exchange returns have been well documented in the past.

Turning our attention to the specification of a set of suitable parameter values for starting the multiprocess model, an ordinary least squares estimation technique was employed on the market model structure (1.1) with the following results being obtained, viz.

$$\alpha = 0,0058 \quad \beta = 0,9424$$

and an unbiased estimate for the variance of e_t being given by $\hat{\sigma}^2 = 0,065$. Accordingly, we set

$$\hat{\mathbf{x}}_0|_0 = \begin{bmatrix} 0,0058 \\ 0,9424 \end{bmatrix}$$

and on noting that

$$\mathcal{E}(\lambda^{-1}) = \delta_0/\rho_0$$

we set $\delta_0 = 10$ and $\rho_0 = 10/0,065$. Setting $\mathbf{C} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ this ensured that large entries were specified for $\text{var}(\lambda) = \delta_0/\rho_0^2$, and $\text{var}(\mathbf{x}_0) = \lambda^{-1}\mathbf{C}_0$, thereby reflecting, to some extent, the uncertainty that we have regarding the accuracy of our starting estimates. The following set of values, $\pi = (0,85; 0,06; 0,07; 0,02)'$, was chosen for the unconditional regime indicator

probability vector. A motivation for the above choice of probability values is given in Smith and West (1983).

For the purposes of making a comparison, an autoregressive market model structure of the form

$$\begin{aligned}r_t &= \alpha + \beta m_t + e_t \\e_t &= \epsilon_t - \psi e_{t-1}\end{aligned}$$

where ϵ_t defines an independent $N(0, \sigma^2)$ random variable, was also computed. A plot of the corresponding prediction residuals, along with their associated forecast variances, is given in figure 3.

5. CONCLUSIONS

Because our sampling period has included the world-wide stock market crash of 1987, the performance of both state space models in dealing with the effects of this type of crash was of interest to the author. An examination of figure 4, where the observed rates of return on the De Beer's stock are plotted against a monthly index value, would seem to indicate that a period of increased volatility in the rate of return on the stock occurred after a monthly index value of about 90: an index value which incidentally corresponds to about the time of the stock market crash in 1987. By dividing the observed data set into a pre- and post stock market crash set of data, a statistical test for the equality of variances of both sets of data confirmed this at the 5% level of significance.

Turning our attention to the performance of the CAPM based state space model structure, an examination of the estimates for the state transition probabilities that are given in figure 1(d) indicates that a steady state evolutionary pattern in the rate of return on De Beer's stock prevailed for most of the early 1980s. This was, however, interspersed (see figure 1(d) model 2) with a periodic change in the level of α_p , that is, the rate of return on the stock that is not related to the rate of return on the market as a

whole. At a time just prior to the 1987 stock market crash, however, a noticeable change in the evolutionary pattern of the rate of return on De Beer's stock seems to have occurred, with the post-stock market crash returns tending to be characterized by a period of high volatility as indicated by the marked increase in the probability of the occurrence of the transient model state.

Turning our attention to the performance of the localized linear growth state space model representation, an examination of the state transition probabilities that are given in figure 2(d) indicates that a steady state evolutionary pattern prevailed for most of the period of the study, and that this was interspersed, periodically, with changes in both the slope (rate of growth) and the level of the return on the stock. Once again, the effect of the 1987 stock market crash on the parameter estimates for μ_i and γ_i is clearly illustrated in figures 2(a) and (b) where a large fluctuation in their estimated values occurred at around the monthly indicator mark of 90.

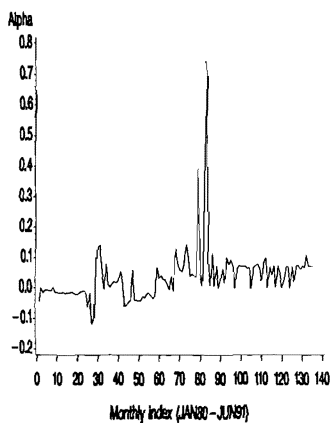
In figures 3(a) and (b) the one-step-ahead forecast errors and variances of both state space model structures are presented. The ability of the CAPM based model structure to adjust to the stock market crash of 1987 is clearly illustrated by the relatively quick manner in which the forecast errors were able to readjust themselves to a more stable pattern soon after the 1987 crash. In contrast, however, the localized linear growth model's prediction residuals appear to be larger in magnitude than those of the CAPM based state space model structure, suggesting that the CAPM based model structure performed better over the time period of the study. In fact, a test for the equality of the one-step-ahead forecast error means was performed for both model structures, with the hypothesis of equality being rejected at the 10% level of significance.

Turning our attention to an examination of the forecasting performance of the autoregressive model structure that is given in figure 3(a), the forecast errors for both the state space based model structures appear to be smaller than those of the autoregressive model. This was further confirmed by conducting a test for the equality of means between the forecast errors of the

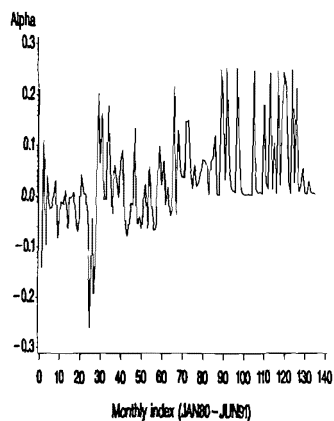
AR model and those of the two state space based model structures, which was rejected at the 5% level of significance. Our analysis can thus conclude that the use of a multiprocess state space model has some merit as a technique for pricing the rate of return on ordinary stock. Further tests need, however, to be carried out on alternative sets of stock price data before one can conclusively prefer the use of a state space based model structure over that of an autoregressive model.

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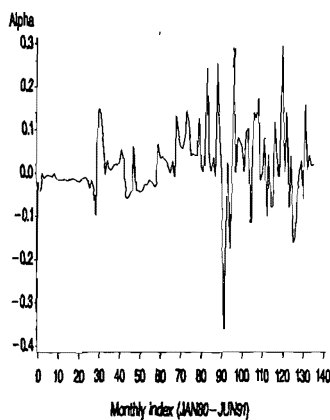
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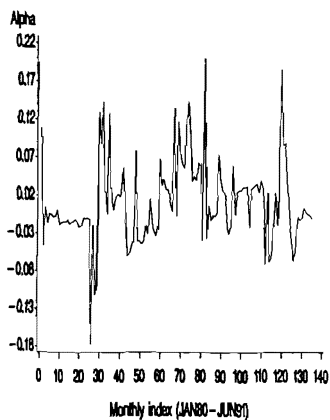
Model regime1



Model regime2

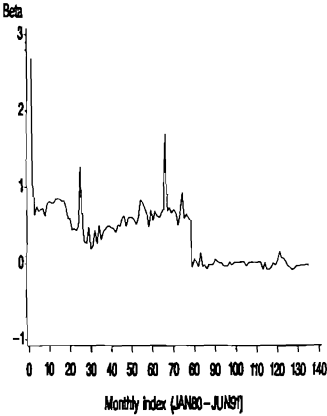


Model regime3

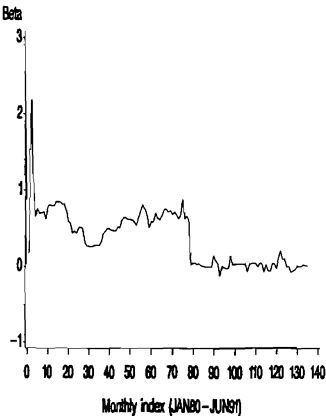


Model regime4

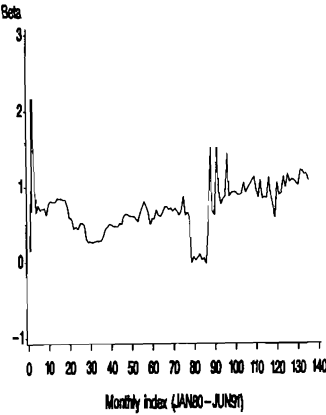
Figure 1(a): Estimates for alpha



Model regime1



Model regime2

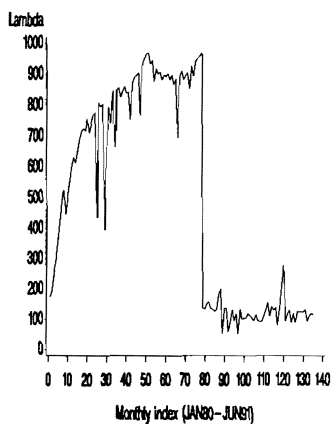


Model regime4

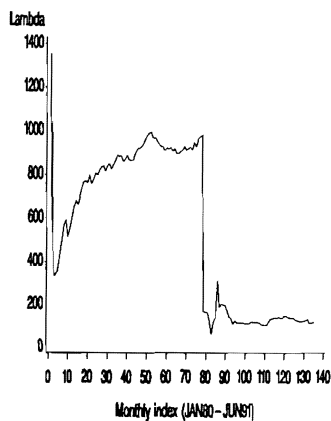


Model regime3

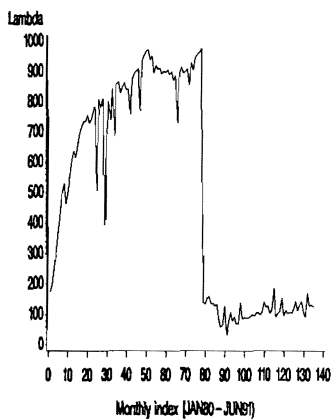
Figure 1(b): Estimates for beta



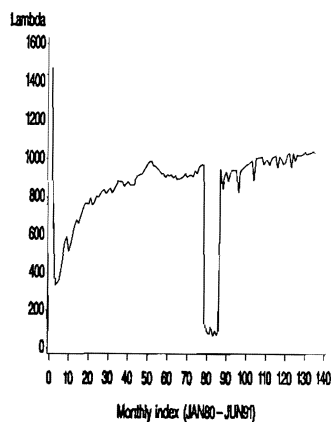
Model regime1



Model regime2



Model regime3



Model regime4

Figure 1(c): Estimates for lambda

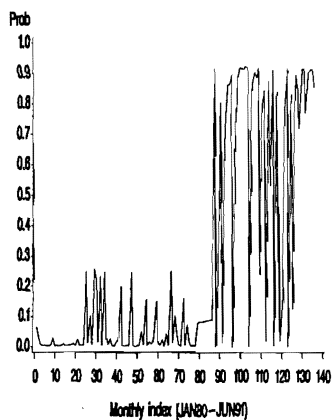
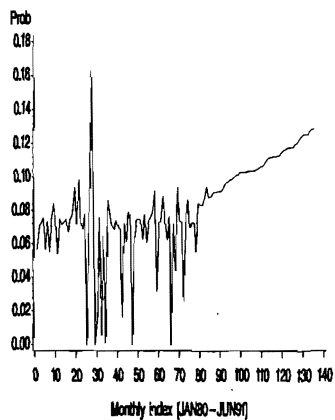
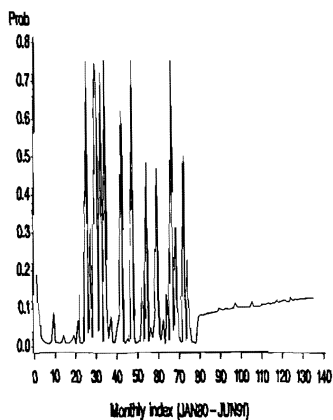
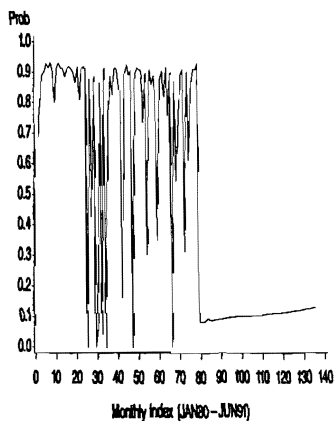
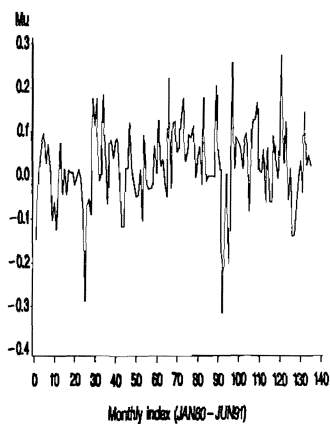
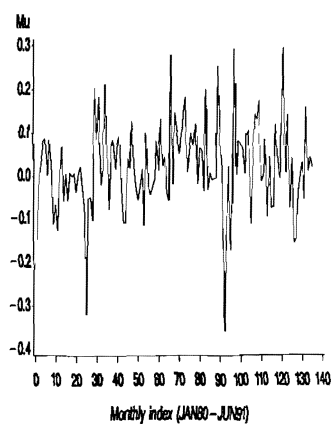


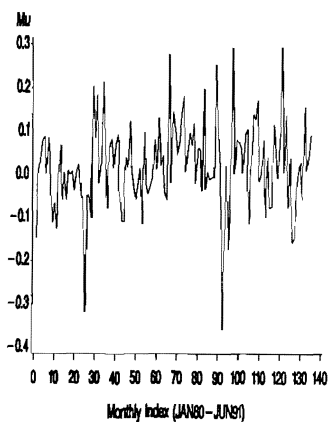
Figure 1(d): Estimates for the regime probability



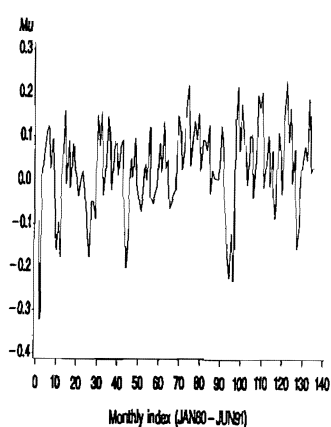
Model regime1



Model regime2

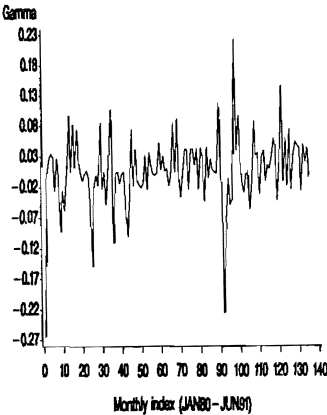


Model regime3

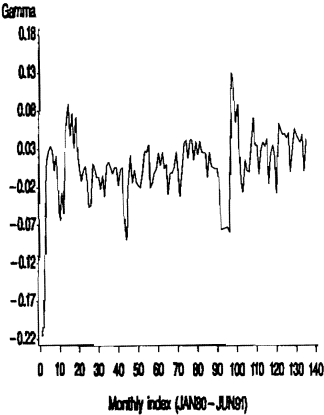


Model regime4

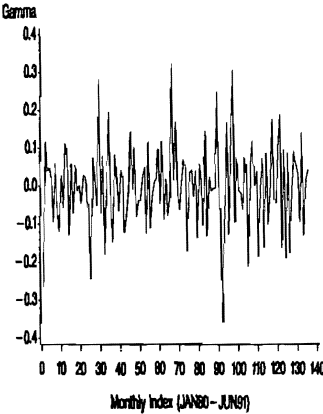
Figure 2(a): Estimates for μ



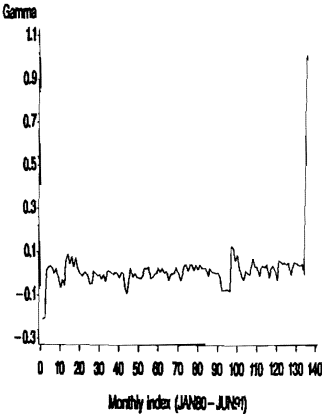
Model regime1



Model regime2

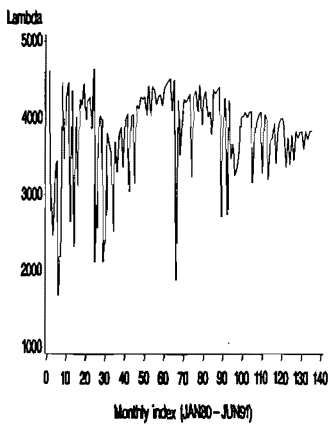


Model regime3

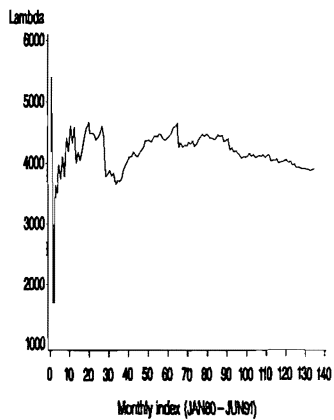


Model regime4

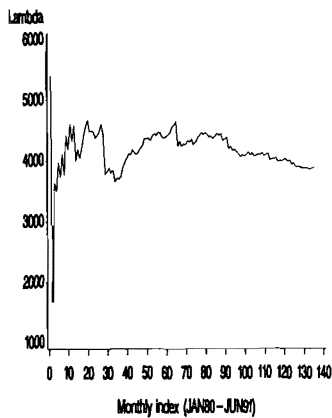
Figure 2(b): Estimates for gamma



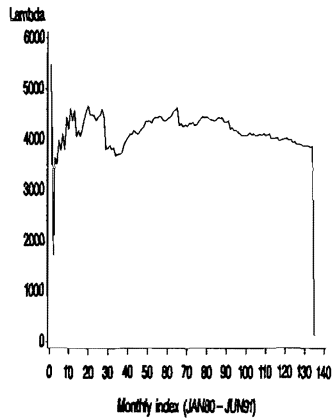
Model regime1



Model regime2

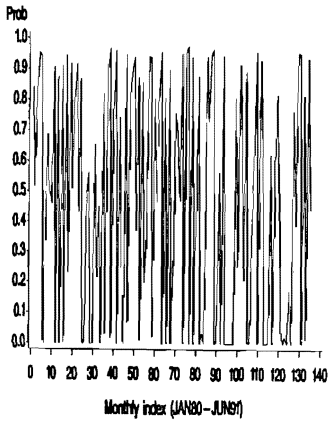


Model regime3

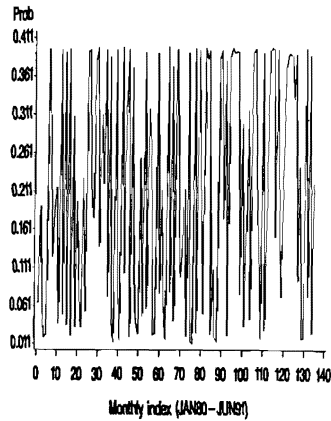


Model regime4

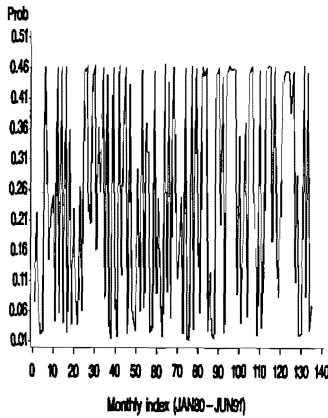
Figure 2(c): Estimates for lambda



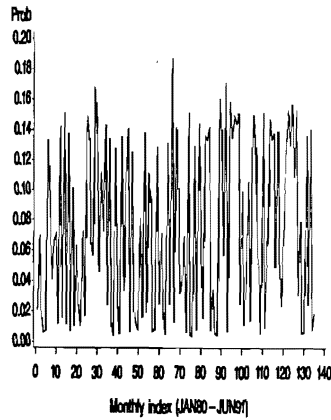
Model 1



Model 2



Model 3



Model 4

Figure 2(d): Estimates for the regime probability

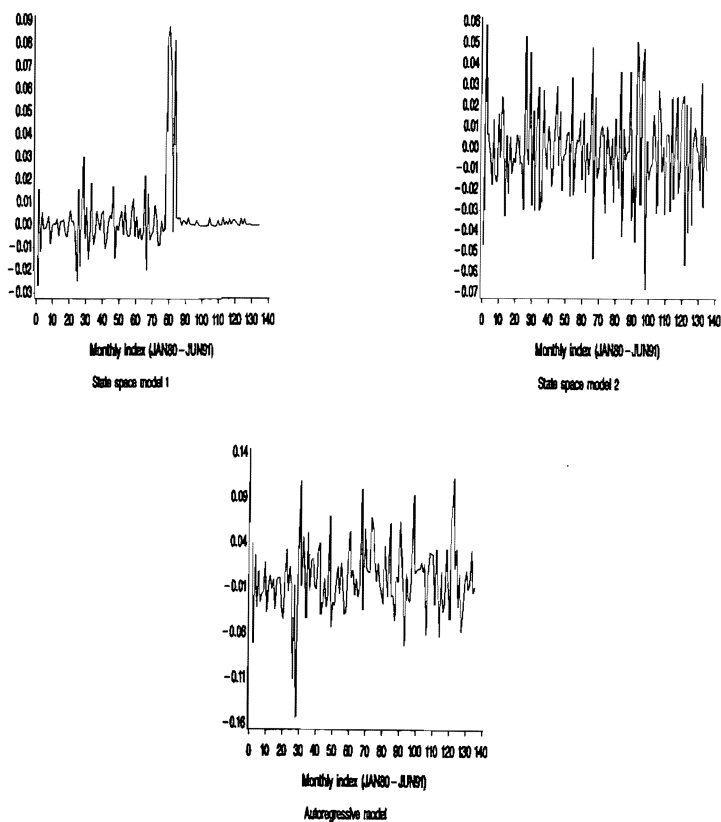


Figure 3(a): The one-step ahead forecast errors

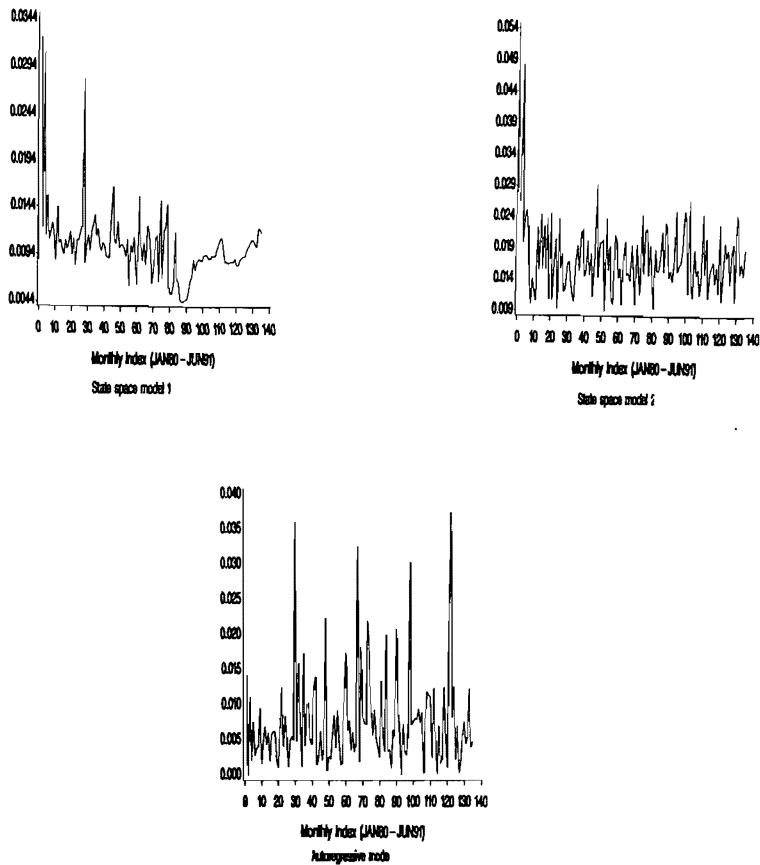


Figure 3(b): The one-step ahead forecast variances

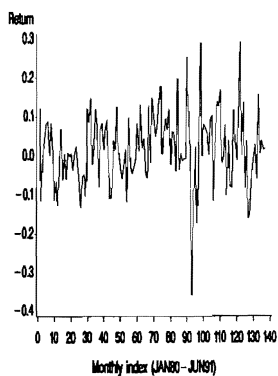


Figure 4: Observed rate of return